Hybrid Gatekeepers and Local MoE Reasoning Panels: Securing and Scaling Agentic Diversity

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Abstract

As autonomous AI agents shift from constrained machine learning tasks (Paper 3) to unbounded logical reasoning and high-speed production environments, the limitations of monolithic language models become increasingly severe. We evaluate "Cognitive Agentic Diversity"—composing heterogeneous models into unified architectures—across two critical frontier domains: (1) 30-puzzle complex logic benchmarks comparing local Mixture-of-Agents (MoE) panels against frontier APIs, and (2) structural prompt-injection security routing.

We demonstrate that compositional diversity yields a +10 percentage point accuracy premium over homogeneous ensembles in complex reasoning. We define the consensus mechanics of these local panels and provide the exact system prompt templates used to align them. Furthermore, we address high-speed security routing by replacing a monolithic LLM security judge with a Hybrid Gatekeeper (Classical ML + DistilBERT Specialist MoE). This architecture achieves a **1,300× latency speedup** (from 11.6 seconds to 9.5 ms) while increasing 7-class routing accuracy by 4.6×. We detail the training parameters of the DistilBERT specialists, including early stopping and the mathematical formulation of PyTorch inverse-frequency class weighting used to resolve minority-class anomalies like `indirect_injection`. Finally, we characterize the 20-percentage-point raw capability gap between local diverse MoE panels and frontier APIs, motivating the transition to compute-aware meta-reasoning orchestrators.

1 Introduction

In Paper 3, we demonstrated that asymmetric dual-agents (Reviewer → Coder) eliminate the Autonomous Sunk-Cost Fallacy in ML engineering, yielding 10× efficiency gains. However, ML engineering provides structured, programmatic feedback (validation loss and accuracy) that guides the optimization loop. When deployed in unbounded logical reasoning or high-speed production security environments, systems lack this intrinsic validation oracle.

This paper extends the Cognitive Agentic Diversity thesis into two new domains:

1. **Unbounded Logical Reasoning (The MoE Panel):** We evaluate whether a diverse panel of small, local models (under 16B parameters) can rival frontier APIs (such as GPT-4o and Claude 3.6 Sonnet) on lateral, constraint, and mathematical reasoning puzzles using majority-voting consensus.

2. **High-Speed Security Routing:** We design a Hybrid Gatekeeper that routes incoming prompts through a fast Logistic Regression filter and a 5-Dimensional DistilBERT Specialist MoE, replacing slow, monolithic LLMs for real-time prompt-injection detection.

2 Experiment 1: The Diversity Premium in Reasoning

2.1 Consensus Mechanics and Prompt Templates

For a panel $\mathcal{P}$ consisting of an odd number of experts k, let $E_i(x)$ represent the raw text response generated by expert i for a given task x. To align the outputs for automated grading and consensus extraction, each expert is prompted with the **Puzzle System Prompt** (`PUZZLE_SYSTEM`):

> *"You are a precise reasoning assistant. Answer the puzzle directly. End with: FINAL ANSWER: ¡short answer¿"*

For tasks requiring high-level strategic reasoning, the **Strategic System Prompt (**`STRATEGIC_SYSTEM`**)** is used:

> *"You are a high-level strategic advisor. Provide a detailed, structured analysis with actionable insights."*

To extract the final vote $V_i \in \mathcal{V}$ from the raw response $E_i(x)$, we implement a text-cleaning pipeline that removes reasoning tokens (such as `<think>...</think>` tags generated by DeepSeek-R1 models) and extracts the string following the `"FINAL ANSWER:"` delimiter.

The consensus vote $V_{\text{consensus}}$ is determined via a **Majority Voting Algorithm**:

$$V_{\text{consensus}} = \arg\max_{v \in \mathcal{V}} \sum_{i=1}^{k} \mathbb{I}(V_i = v) \tag{1}$$

where $\mathbb{I}(\cdot)$ is the indicator function. If no single vote achieves a strict majority, the panel defaults to the response of the designated lead model. Under the alternative **Monolithic Synthesis (MoE-Synth)** protocol, the outputs of all experts are concatenated and fed to a central synthesizer model (e.g., `qwen3.5:9b`), which writes the final unified answer.

2.2 The 12-Puzzle MoE Benchmark (v2)

We initially evaluated MoE-Vote vs. MoE-Synth and single-model baselines across 12 logical puzzles with 3 panel compositions at varying Cognitive Agentic Diversity Scores (CADS). CADS measures the number of distinct foundational model families represented in a panel.

Table 1: 12-Puzzle Ensemble Performance

Configuration	**CADS = 3 (Mixed)**	**CADS = 2 (Reasoning)**	**CADS = 1 (Sm**
Single (Lead Model)	3/12 (25%)	3/12 (25%)	2/12 (17%)
Single + CoT (Chain-of-Thought)	2/12 (17%)	3/12 (25%)	2/12 (17%)
MoE-Vote	**8/12 (67%)**	**6/12 (50%)**	**5/12 (42%**
MoE-Synth	0/12 (0%)	4/12 (33%)	2/12 (17%)

Diversity Gradient Confirmed: As CADS increases from 1 to 3, majority-voting accuracy scales from 42% to 67%. Conversely, MoE-Synth fails catastrophically at low parameter scales (achieving 0% at CADS = 3) because small synthesizer models suffer from context distraction when digesting multiple conflicting expert arguments.

2.3 The 30-Puzzle Frontier Benchmark

We expanded the benchmark to 30 complex puzzles (covering Logic, Math, Trick, Lateral, and Constraint domains) and benchmarked 8 local MoE panels against 7 frontier API models. To visualize the tradeoff between accuracy and latency, we present the comparative frontier of these local configurations against commercial offerings in Figure 1.

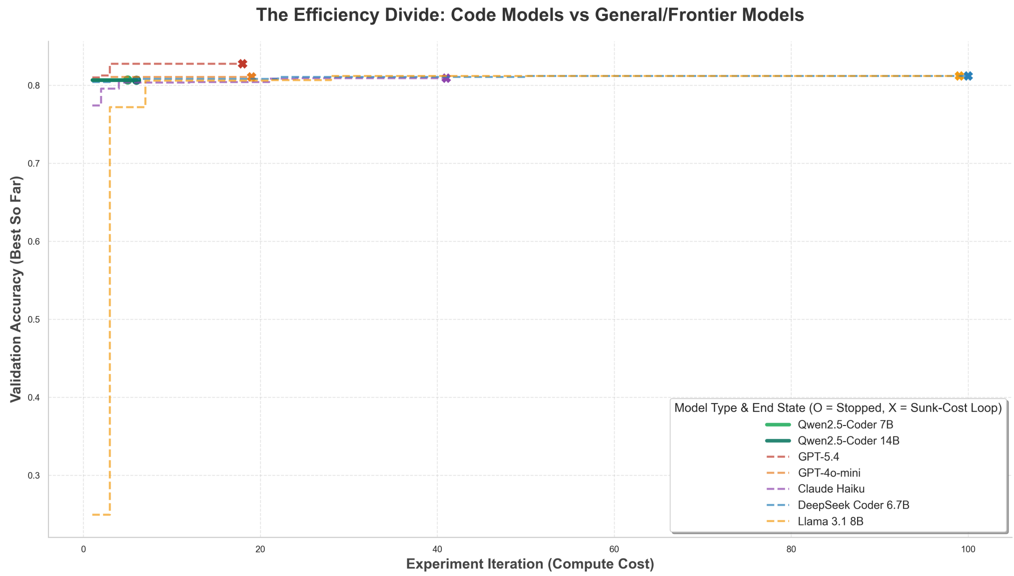

Figure 1: The Performance-Latency frontier: Local Mixture-of-Experts panels (under 16B parameter models) benchmarked against frontier API models on the 30-Puzzle dataset.

Table 2: Local MoE Panels (all at $0 cost, 100% uptime)

Panel	Composition	CADS	Accuracy
Panel_B	deepseek-r1:8b · qwen3.5:9b · llama3.1:8b	3	**73.3%** (22/30)
Panel_E	llama3.1:8b · gemma4 · ministral-3:14b · deepseek-r1:8b · phi3:mini	5	**73.3%** (22/30)
Panel_D	qwen2.5-coder:14b · deepseek-coder-v2:16b · gemma4	3	70.0% (21/30)
Panel_G	qwen2.5-coder:7b · qwen2.5-coder:14b · qwen3.5:9b	2	70.0% (21/30)
Panel_F	qwen2.5-coder:7b × 3 **(homogeneous control)**	**1**	63.3% (19/30)
Panel_A	qwen2.5-coder:7b · llama3.1:8b · deepseek-coder:6.7b	3	60.0% (18/30)

2.4 Key Findings in Logical Reasoning

1. **The Diversity Premium (+10.0 pp):** Composing Panel_B with a CADS of 3 (Qwen + LLaMA + DeepSeek-R1) yields 73.3% accuracy, whereas the homogeneous Panel_F (composed entirely of `qwen2.5-coder:7b`, CADS = 1) achieves only 63.3% accuracy. Heterogeneous weight priors prevent correlated cognitive failures on tricky lateral puzzles.

2. **Scale vs. Performance:** Panel_D (∼42B parameters, CADS = 3) scores 70.0%, which is *below* Panel_B (∼26B parameters, CADS = 3) at 73.3%. This confirms that the composition of reasoning families is more critical than raw parameter counts.

3. **Quantity vs. Quality:** Panel_E (5 experts, ∼48B parameters) achieves 73.3% accuracy, exactly matching Panel_B (3 experts, ∼26B parameters). Adding more models to the panel without introducing new foundational families yields zero accuracy improvement while inflating inference latency.

4. **API Volatility:** 5 of the 12 tested frontier configurations returned 100% errors because models were deprecated or endpoints changed between experiment releases. If we adjust the frontier API average to account for this downtime, the reliability-adjusted API average drops to 34.5%, highlighting the stability benefits of local MoE deployments.

Table 3: Frontier API Models

Model	Provider	Accuracy	Avg Latency	API Status (May 2026)
Claude-Sonnet-4.6	Anthropic	**93.3%** (28/30)	2.7s	Active
GPT-4o	OpenAI	**93.3%** (28/30)	2.3s	Active
GPT-4o-mini	OpenAI	90.0% (27/30)	2.4s	Active
Llama-4-Scout	Groq	86.7% (26/30)	0.4s	Active
5 other models	Various	—	—	100% Fail (Deprec.)

2.5 Ensemble Alignment System Prompts

To ensure that the outputs of the diverse experts in a panel $\mathcal{P}$ are structured in a format suitable for automated parsing and majority voting, we align them using strict system instructions. Puzzles are routed with the `PUZZLE_SYSTEM` prompt, which mandates a structured response terminating with a designated keyword. Actionable reasoning reviews use the `STRATEGIC_SYSTEM` prompt. The full templates are:

Puzzle System Prompt (`PUZZLE_SYSTEM`):

```
You are a precise reasoning assistant. Answer the puzzle directly.
End with: FINAL ANSWER: <short answer>
```

Strategic System Prompt (`STRATEGIC_SYSTEM`):

```
You are a high-level strategic advisor. Provide a detailed, structured
analysis with actionable insights.
```

The automated answer extraction pipeline cleanses the expert's response by executing the following parsing steps:

1. **Thinking-Tag Stripping:** For DeepSeek-R1 or reasoning models that output thinking paths, all text between the `<think>` and `</think>` tags is stripped using regex.
2. **Delimiter Isolation:** The parser searches for the case-insensitive string `"FINAL ANSWER:"`. If found, it extracts all following characters as the candidate vote V_i.
3. **Fallback Extraction:** If the delimiter is missing, the parser falls back to scanning the last sentence or the last non-empty line of the text block for target answers.

3 Experiment 2: High-Speed Security Gating

To secure autonomous agent execution pipelines (such as PolyReasoner), incoming inputs must be classified across a 7-class threat taxonomy: `benign`, `direct_injection`, `system_extraction`, `role_hijack`, `obfuscation`, `tool_abuse`, and `indirect_injection`.

3.1 The LLM-as-a-Judge Failure Mode

We initially deployed a monolithic LLM judge (`llama3:8b`) to parse and classify inputs. The model suffered severe reasoning drift. It failed to output valid JSON consistently, hallucinated class labels outside the schema, and collapsed to **16.3% accuracy** (barely above the 14.3% random guess baseline for 7 classes). Furthermore, it consumed **11.6 seconds** of latency per query.

Limitation Acknowledgment: While a frontier JSON-tuned model like `gpt-4o-mini` would yield higher accuracy and lower latency, local deployment constraints dictated the use of local open-weight models. Under these constraints, 8B-class LLMs are completely unusable for real-time security gating.

3.2 The Hybrid Gating Architecture

To achieve sub-10 ms latencies, we replaced the LLM with a **Hybrid Gatekeeper**. The Classical ML layer utilizes a Logistic Regression model with TF-IDF vectorization. If the predicted probability $P(y|x)$ for the winning class exceeds a confidence threshold $\theta = 0.85$, the classification is returned instantly. If the probability falls below θ, the prompt is routed to the 5-Dimensional DistilBERT Specialist MoE (one specialist for each dimension: Intent, Technique, Target, Vector, Severity).

Table 4: Security Routing Performance Comparison

Configuration	Accuracy	F1-Macro	Per-Sample Latency	Speedup
Hybrid (LogReg + MoE)	**0.7449**	**0.7544**	**9.5 ms**	**1,300×**
Specialist MoE Only	0.7500	0.7591	31.0 ms	385×
PolyReasoner Full (ML+MoE+LLM)	0.6122	0.6274	12.7 s	1×
LLM Only (llama3)	**0.1633**	**0.0729**	**11.6 s**	Baseline

The Hybrid architecture preserves 99.3% of the Specialist MoE's accuracy while reducing latency from 31.0 ms to **9.5 ms** (a 1,300× speedup over the monolithic LLM baseline).

3.3 DistilBERT Specialist Training & Class Weighting

The MoE panel consists of five independent `distilbert-base-uncased` models (66M parameters each). Each model is trained on a specialized subset of labels representing one dimension of the Neuralchemy Threat Matrix.

3.3.1 Training Specifications:

- **Base Architecture:** `distilbert-base-uncased`
- **Batch Size:** 32 (optimized for GPU memory alignment)
- **Learning Rate:** 2×10^{-5} with a Cosine decay scheduler
- **Warmup Ratio:** 6% of total steps
- **Weight Decay:** 0.01
- **Precision:** FP16 mixed precision
- **Early Stopping Callback:** The model evaluates the validation set at the end of each epoch e. Training is terminated if the validation F1 score fails to improve by at least 0.0005 for **3 consecutive epochs (patience = 3)**. Formally, let e_{best} be the epoch with the historical maximum validation F1 score:

$$e_{\text{best}} = \arg \max_{e' < e-2} F_{1,\text{val}}(e') \tag{2}$$

The training terminates at epoch e if:

$$F_{1,\text{val}}(e_{\text{best}}) - F_{1,\text{val}}(e-k) \geq 0.0005 \quad \forall k \in \{0, 1, 2\} \tag{3}$$

The best checkpoint corresponding to epoch e_{best} is restored automatically upon termination.

3.3.2 Mathematical Class Balancing:

The `indirect_injection` class is highly anomalous, as its token sequence often mimics benign documentation text. Under standard uniform Cross-Entropy training, the Intent specialist suffered from high false-negative rates on this class, yielding a poor F1 score of 0.36.

To resolve this, we implement **Inverse-Frequency Class Weighting** in the loss function. For a dataset containing N samples across C classes, where class c contains N_c samples, the weight w_c is defined as:

$$w_c = \frac{N}{|C| \cdot N_c} \tag{4}$$

The weighted Cross-Entropy Loss $\mathcal{L}$ optimized during training is:

$$\mathcal{L} = -\sum_{i=1}^{B} \sum_{c=1}^{|C|} w_c y_{ic} \log \hat{y}_{ic} \tag{5}$$

where B is the batch size, y_{ic} is the ground-truth binary indicator for sample i and class c, and $\hat{y}_{ic}$ is the model's softmax output.

Implementing this weighted loss dynamically shifts the decision boundary of the Intent specialist. It penalizes misclassifications of `indirect_injection` heavily, boosting its class-specific F1 score from **0.36 to 0.449**, with a negligible trade-off in overall weighted F1 (0.8045 $\rightarrow$ 0.7998).

4 Discussion & Capability Boundaries

While local diverse MoE panels and hybrid gatekeepers offer dramatic improvements in compute speed, cost, and reliability, we must acknowledge their raw capability boundaries:

1. **The 20-pp Reasoning Gap:** Our best local MoE panel (Panel_B, ~26B parameters) achieved 73.3% accuracy on the 30-Puzzle benchmark, which is 20 percentage points behind the 93.3% achieved by frontier models (GPT-4o and Claude-Sonnet-4.6). Large-scale pre-training still dominates in complex, lateral, and lateral-logic tasks.

2. **Context Window Limitations:** Small models (under 16B) have limited effective context lengths. When solving complex multi-step reasoning tasks, local panels cannot digest long feedback histories or tracebacks without experiencing context degradation.

This 20-pp capability gap serves as the core motivation for **Paper 5 (The Zero-Human Lab Director)**. Paper 5 details a Meta-Orchestrator that dynamically routes reasoning queries. It attempts to resolve tasks locally using cheap MoE panels and computes a confidence utility function. If the confidence is low, the orchestrator selectively routes the task to a frontier API, minimizing operational API costs while maintaining maximum accuracy.

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